

THEOREM. *In an integral domain $\mathbf{A}$, for every polynomial P in the variable x with coefficients in $\mathbf{A}$, $P \in \mathbf{A}[x]$, and every $r \in \mathbf{A}$, there exist unique $\mathcal{Q} \in \mathbf{A}[x]$ and $\mathcal{R} \in \mathbf{A}$ such that*

$$P(x) = (x - r) \mathcal{Q}(x) + \mathcal{R},$$

where

$$\mathcal{R} = P(r).$$

○ **1. The existence**

Let $P(x)$ be a polynomial in $\mathbf{A}[x]$ of degree n ,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i,$$

where each coefficient $a_i \in \mathbf{A}$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^n a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i=0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned} \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i &= (a_0 - a_0) + \sum_{i=1}^n (a_i x^i - a_i r^i) \\ &= \sum_{i=1}^n a_i (x^i - r^i). \end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i -th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

LEMMA. For any integer $n \geq 1$, the difference of two n -th powers, $x^n - c^n$, can be factored by $(x - c)$. That is, there exists a polynomial $Q_{n-1}(x, c)$ such that

$$x^n - c^n = (x - c)Q_{n-1}(x, c).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x - c)(1),$$

where the constant $1 = Q_0(x, c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $Q_{k-1}(x, c)$ such that

$$x^k - c^k = (x - c)Q_{k-1}(x, c).$$

Induction Step. To show that the lemma holds for $k + 1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^kx$ and grouping as

$$x^{k+1} - xc^k + c^kx - c^{k+1} = (x^{k+1} - xc^k) + (c^kx - c^{k+1}).$$

Factoring out the common components from each group gives

$$x(x^k - c^k) + c^k(x - c).$$

Applying the inductive hypothesis to substitute $(x - c)Q_{k-1}(x, c)$ for $(x^k - c^k)$ yields

$$x(x - c)Q_{k-1}(x, c) + c^k(x - c).$$

Factoring out the common multiplier $(x - c)$ results in

$$(x - c)(xQ_{k-1}(x, c) + c^k),$$

where $xQ_{k-1}(x, c) + c^k = Q_k(x, c)$ is a polynomial — since $Q_{k-1}(x, c)$ is a polynomial by the inductive hypothesis, and polynomial rings are closed under multiplication and addition.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $Q(x, c)$ polynomials are

$$\begin{aligned}
1 &= Q_0 \\
x(1) + c^1 &= x + c &= Q_1 \\
x(x + c) + c^2 &= x^2 + xc + c^2 &= Q_2 \\
x(x^2 + xc + c^2) + c^3 &= x^3 + x^2c + xc^2 + c^3 &= Q_3 \\
x(x^3 + x^2c + xc^2 + c^3) + c^4 &= x^4 + x^3c + x^2c^2 + xc^3 + c^4 &= Q_4 \\
&\dots \\
x(x^{n-2} + x^{n-3}c + \dots + xc^{n-3} + c^{n-2}) + c^{n-1} &= x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1} = Q_{n-1} \\
x(x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1}) + c^n &= x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = Q_n
\end{aligned}$$

In more compact notation with the summation symbol

$$Q_n(x, c) = x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = \sum_{i=0}^n x^{n-i}c^i$$

and

$$xQ_{n-1}(x, c) + c^n = x \sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^n x^{n-i}c^i = Q_n(x, c).$$

The exact factorization is
$$x^n - c^n = (x - c) \sum_{i=0}^{n-1} x^{n-1-i}c^i.$$

Returning to
$$P(x) - P(r) = \sum_{i=1}^n a_i (x^i - r^i),$$

the lemma yields*

$$x^i - r^i = (x - r)Q_{i-1}(x, r)$$

for every $i \geq 1$. Substituting these factorizations term-by-term gives

$$P(x) - P(r) = \sum_{i=1}^n a_i (x - r)Q_{i-1}(x, r) = (x - r) \sum_{i=1}^n a_i Q_{i-1}(x, r).$$

Since each $Q_{i-1}(x, r)$ is a polynomial in $\mathbf{A}[x]$ and polynomial sums are polynomials, $\sum_{i=1}^n a_i Q_{i-1}(x, r)$ belongs to the same polynomial ring $\mathbf{A}[x]$. Define

$$\sum_{i=1}^n a_i Q_{i-1}(x, r) = \mathcal{Q}(x) \in \mathbf{A}[x].$$

Hence
$$P(x) - P(r) = (x - r)\mathcal{Q}(x)$$

* From a deeper algebraic viewpoint, $Q_{i-1}(x, c)$ from the lemma belongs to the polynomial ring $\mathbf{A}[x, c]$. Replacing the indeterminate c by the element $r \in \mathbf{A}$ is the evaluation homomorphism $\mathbf{A}[x, c] \rightarrow \mathbf{A}[x]$. Thus $Q_{i-1}(x, r)$ is a polynomial in $\mathbf{A}[x]$.

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or equivalently $P(x) = (x - r)Q(x) + P(r)$.

The latter exhibits the required representation with $\mathcal{R} = P(r) \in \mathbf{A}$. Ergo there exist $Q \in \mathbf{A}[x]$ and $\mathcal{R} \in \mathbf{A}$. ●

○ **2. The uniqueness**

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